Inorganic cume

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Polytopal complexes

Complexes having the same ligand set but different overall geometry are referred to as ***polytopal isomers***. One example are the $[Ni(CN)_5]^{3-}$ anions of the salt $[Cr(en)_3][Ni(CN)_5]$, which exist as a 1:1 mixture of trigonal bipyramid (***tbp***, D_{3h}) and square pyramid (***sp***, C_{4v}) isomeric anions in the solid state.

1) What is the maximum number of IR bands you could observe in the C≡N stretching region for solid $[Cr(en)_3][Ni(CN)_5]$? Assume that no "unexpected" bands occur due to solid state effects and that there are no accidental degeneracies.

Remember, the selection rules for vibrational spectra (IR and Raman) can be derived using the character table of the point group for a species.

2) The compound $Fe(CO)_5$ shows an IR in the CO stretching region with two bands and a Raman with three bands (two are polarized). One of the vibrational bands is observed in both the IR and Raman spectrum; however, the polarized Raman bands are not observed in the IR. Is the data most consistent with ***tbp*** or ***sp*** geometry? Show proof for your assertion.

Totally polarized Raman bands are associated with vibrations that belong to the total symmetric representation for the point group of a molecule.

3.) The ^{14}C magnetic resonance spectrum of $Fe(CO)_5$ consists of a single resonance peak (i.e. all the CO's are equivalent). Can this be consistent with data in question #2? Justify your answer.

D_{3h} ($\bar{6}m2$)	E	$2C_3$	$3C_2$	σ_h	$2S_3$	$3\sigma_v$		
A_1'	1	1	1	1	1	1		x^2+y^2, z^2
A_2'	1	1	−1	1	1	−1	R_z	
E'	2	−1	0	2	−1	0	(x, y)	(x^2-y^2, xy)
A_1''	1	1	1	−1	−1	−1		
A_2''	1	1	−1	−1	−1	1	z	
E''	2	−1	0	−2	1	0	(R_x, R_y)	(xz, yz)

C_{4v} ($4mm$)	E	$2C_4$	C_2	$2\sigma_v$	$2\sigma_d$		
A_1	1	1	1	1	1	z	x^2+y^2, z^2
A_2	1	1	1	−1	−1	R_z	
B_1	1	−1	1	1	−1		x^2-y^2
B_2	1	−1	1	−1	1		xy
E	2	0	−2	0	0	$(x, y)(R_x, R_y)$	(xz, yz)